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DockTGrid: A Python-Based Tool for Generating Deep Learning-Ready Voxel Grids of Molecular Complexes

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Summary

Deep learning (DL) techniques are at the forefront of advancements in molecular modelling tools, significantly impacting the field of computer-aided drug design (CADD) (Chen et al., 2018; Guedes et al., 2018; Shen et al., 2020; Zhang et al., 2017). In this context, voxel grid representations have emerged as a straigthforward way to depict the three-dimensional molecular structures of protein-ligand complexes. These representations are used as input to DL models, such as convolutional neural networks (CNNs), to make inferences on a range of CADD tasks, including molecular docking (McNutt et al., 2021; Ragoza et al., 2017), binding affinity prediction (Jiménez et al., 2018; Li et al., 2019; Liu et al., 2021; Ragoza et al., 2017; Stepniewska-Dziubinska et al., 2018), binding site prediction (Jiménez et al., 2017; Liang & Jacobson, 2022), molecular generation (Ragoza et al., 2022) and virtual screening (Ragoza et al., 2017; Sunseri & Koes, 2021). In this work, we introduce a Python-based software, `DockTGrid`, which provides a set of tools to generate, customize and manipulate voxel grid representations of protein-ligand complexes ready to be used as input to DL models.

Voxel grids approach the challenge of molecular representation through a computer vision lens. A voxel (*volume element*) corresponds to a value on a regular grid in the three-dimensional space. A voxel grid is a 4-dimensional tensor with dimensions $c \times w \times h \times d$, where w, h, and d are the grid's width, height, and depth, respectively, and c is the number of channels (similar to RGB channels in 2D image representations). Voxel grids effectively capture the 3D spatial relationships between atoms and their local environment, making them ideal inputs for CNNs equipped 3D convolutional layers.

The values each voxel holds are defined by the occupancy model used. Space-filling occupancy models, which lead to denser representations and thereby reduce unutilized space within the grid, are frequently employed. These models incorporate the van der Waals radius of atoms and employ a pair correlation function to depict the electronic surface in space. `DockTGrid` includes a space-filling occupancy model (Jiménez et al., 2017), which is mathematically expressed as:

$$v(d) = 1 - \exp(-(\frac{r_{vdw}}{d})^{12}).$$

Here, $v(d)$ represents the voxel occupancy value at a distance d from the atom's center, with r_{vdw} denoting the van der Waals radius of the atom.

In CADD and molecular contexts, grid channels can represent various physicochemical properties of atoms. For instance, channels may be designated to reflect specific atomic properties such as

hydrophobicity, hydrogen bond acceptance or donation, ionizability (either positive or negative), partial atomic charges, and the presence of metals (Jiménez et al., 2018; Stepniewska-Dziubinska et al., 2018). Alternatively, channels might represent different chemical elements, including carbon, hydrogen, oxygen, and nitrogen (Li et al., 2019; Ragoza et al., 2022). Figure 1 demonstrates how channels for carbon and hydrogen atoms in a ligand are depicted using the space-filling occupancy model previously described.

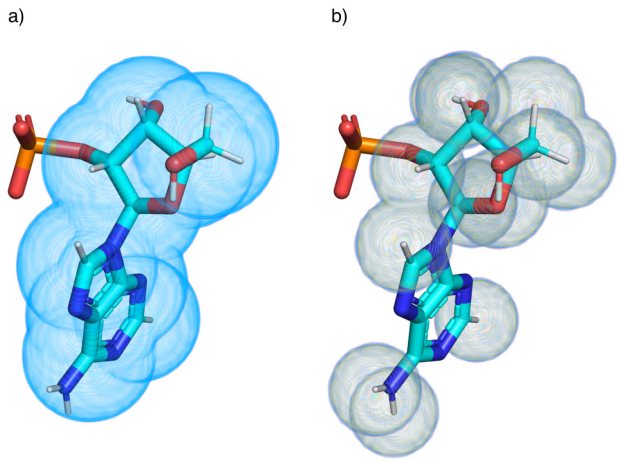

Figure 1: Voxel-Based Representation of Carbon and Hydrogen Atoms Generated with `DockTGrid`. In this visualization, carbon (a) and hydrogen (b) atoms within the adenosine 2'-monophosphate molecule (PDB ID: 6RNT) are depicted using a space-filling occupancy model across two distinct channels.

Statement of need

Related software packages — available on public package repositories (e.g., PyPI, for Python) — offer capabilities for generating voxel grid representations of molecular data tailored for DL applications. Notable examples include `libmolgrid` (Sunseri & Koes, 2020), `moleculekit` (Doerr et al., 2016), and `deepchem` (Ramsundar et al., 2019). However, users may encounter challenges with these packages in feature customization or programming language diversity. In some of these, not all functionalities are developed in Python; incorporating programming languages like C++, which could limit accessibility for users primarily familiar with Python in the DL community.

`DockTGrid` was developed entirely in Python 3 and is built atop the widely-used DL library `PyTorch`. This foundation enables seamless integration and rapid, GPU-accelerated grid generation. The software offers customizable grid channels through a straightforward and user-friendly object-oriented interface, as illustrated in the software documentation (available at https://docktgrid.readthedocs.io). Designed for use by researchers and DL practitioners in the CADD field, `DockTGrid` presents an accessible and open-source solution for generating voxel grid representations of protein-ligand complexes.

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Conflict of Interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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